Problem 09: Ancestral Queries

Time limit: 1 second

The Khan family submitted their digital family tree to the Lahore Heritage Society. The judges were impressed by how efficiently the family had documented their ancestry using algorithmic thinking. They even asked Zara to present her solution at the Lahore Tech Expo, where it became a hit among historians and tech enthusiasts alike.

The Khan family tree looked like this:

- **Dada Abu**: The patriarch of the family.
  - His children: **Ali** and **Fatima**.
    - Ali's children: **Hassan** and **Ayesha**.
      - Hassan's children: **Bilal** and **Sana**.
        - Bilal's children: **Amna** and **Usman**.
    - Fatima's children: **Zainab** and **Omar**.
      - Zainab's children: **Leena** and **Yusuf**.
      - Omar's children: **Hania** and **Saad**.

With the digital family tree in place, the Khan family organized a **grand reunion** at their ancestral home in Lahore's **Old City**. Family members from all over Pakistan gathered to celebrate their shared history and learn more about their ancestors.

- **Amna** discovered that her 4th ancestor was **Dada Abu**, the patriarch of the family.
- **Leena** learned that her 2nd ancestor was **Fatima**, her great-grandmother.
- **Saad** was amazed to find out that his 3rd ancestor was **Ali**, a prominent businessman in Lahore's history.

The Khan family's story became a symbol of **unity, heritage, and innovation** in Lahore. Their digital family tree not only helped them connect with their past but also inspired other families in the city to document their own histories using technology. As Dada Abu often said, **"A family's roots are like the foundations of a building. The stronger they are, the taller we can rise."**

The story highlights the importance of family, heritage, and technology in preserving our connections to the past. Just like the binary lifting technique helps us efficiently find ancestors, understanding our roots helps us navigate the present and future with clarity and purpose.

Description:

Given a family tree of n members and q queries, for each query, find the k-th ancestor of a given family member. If the k-th ancestor does not exist, return -1.

Input:

- The first line contains the number of testcases (t).
- The second line contains n (number of family members), q (number of queries) and r (root of tree).
- The next $n-1$ lines describe the parent-child relationships in the family tree.
- The next q lines contain pairs (u, k), where u is the family member and k is the ancestor level.
- All numbers in a line are single space separated.

Output:

- For each query, print the k-th ancestor of u. If it doesn't exist, print **-1**.

Input	Output
1 5 2 5 5 3 1 3 4 3 1 2 4 2 1 3	5 -1